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NCC Global Assignment

NCC education

Qualification: L5DC-BM

Unit Name: Principles of Business Operations

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Assignment Marking Sheet

Awarding Great British Qualifications

Centre Name	Talent Nurtures Educational Institute
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NCC Education Candidate ID No.	213142
Examination Date	15/8/2025
Examination Unit Title	Principles of Business Operations

Questions answered by the candidate:

Please insert the question numbers you have answered in the table below.

Question number attempted (To be completed by Candidate)	*Marks Awarded (To be completed by 1st marker)*	*Marks Awarded (To be completed by 2nd marker)*
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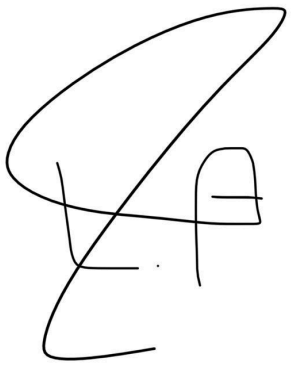
I confirm the following details:

Candidate Name:	Mhd Laith Alkhateeb
Candidate ID Number:	213142
Qualification:	L5DC-BM
Unit:	Principles of Business Operations
Centre:	Talent Nurtures Educational Institute
Word Count:	2860

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I confirm that I have not exceeded the stipulated word limit by more than 10%.

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A comprehensive overview of a technology company's current business operations

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Overview

Microsoft Corporation is a leading American multinational technology company, renowned for its significant impact on the computer software industry. Founded in the mid-1970s by Bill Gates and Paul Allen, Microsoft began as a small venture focused on developing programming languages, eventually rising to prominence with its operating systems and productivity software, such as Windows, Word, Excel, and PowerPoint. By the early 1990s, Microsoft had established itself as the dominant player in the market, controlling approximately 90% of the computer operating systems worldwide.

Despite facing legal challenges regarding its market position, the company successfully navigated these hurdles and continued to innovate. Over the years, Microsoft expanded into various sectors, including video gaming with the Xbox console, internet services with Bing, and cloud computing through Azure. In recent years, Microsoft has embraced developments in artificial intelligence, culminating in strategic partnerships and acquisitions, such as the purchase of video game publisher Activision Blizzard.

Today, Microsoft remains one of the most valuable companies globally, generating substantial revenue and employing tens of thousands of people. It continues to shape the technological landscape, although it faces competition in the evolving realms of mobile and cloud technologies. (ebsco.com , 2024)

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Task 1

Microsoft Value Chain Analysis

1- Inbound Logistics:

Purchasing parts and software development inputs is the main focus of Microsoft's inbound logistics. It keeps an international network of suppliers for hardware parts used in devices like the Xbox and Surface. Microsoft only collaborates with suppliers who follow its ethical sourcing policies, enforcing strict quality and sustainability criteria. Reducing packaging and using more recycled materials are examples of environmental initiatives. In order to optimize code-sharing and version control, the company uses just-in-time delivery for manufacturing and centralizes digital inputs through GitHub and Azure. This strategy ensures continuity and innovation across product lines while increasing efficiency and lowering time-to-market.

2- Operations:

Software development, cloud infrastructure management, and device production are among Microsoft's operations. These attributes characterize the company's strength as a leader in the SaaS business model. The business iterates on Windows, Microsoft 365, and Azure using agile development approaches. Security patches and upgrades are streamlined by DevOps and lean methodologies. Automation and sophisticated robots are used by hardware operations in contract manufacturing to increase output while upholding high standards. With cloud infrastructure spread across more than 60 countries, scalable and reliable services are made possible.

3- Outbound Logistics:

For the majority of its goods, including Office installations and Windows upgrades, Microsoft's outbound logistics rely on digital distribution. Retail partners and e-commerce platforms are part of a global logistical network that distributes physical items, such as Xbox consoles and Surface devices. Software deployment is accelerated globally by its Azure content delivery network, and delivery costs and timelines are decreased through integrated supply chain planning.

4- Marketing and Sales:

Microsoft has a multichannel approach that includes enterprise sales teams, partner ecosystems, and digital advertising. Campaigns frequently emphasize innovation, productivity, and security, particularly for Azure and Microsoft 365. Its reach is expanded by strategic alliances with businesses and educational institutions. Transparent communication, ongoing product enhancements, and packaged service offers that increase value and stickiness all help to build brand loyalty.

5- Service:

Microsoft offers comprehensive post-purchase assistance via chatbots, online help centers, and enterprise level premium support. For Azure subscribers, services include proactive system monitoring, technical help, and software tutorials. Long term client pleasure is guaranteed by loyalty programs, subscription perks, and regular updates. Community support is provided by Xbox Live and Microsoft 365, which improves user retention and engagement.

Supporting Activities

1- Firm infrastructure:

Microsoft's strong governance, financial planning, and compliance processes support its worldwide operations. To effectively handle its various product lines, the corporation uses a matrix organizational structure. Transparency, sustainability, and ethical AI development are prioritized by the leadership, and its strategic planning teams are always looking for new growth prospects. Scale and sophisticated financial forecasting models enhance cost management.

2- Human Resources Management

Attracting, nurturing, and keeping great talent is the main goal of Microsoft's HR strategy. this goal is also examined in its CSR programs, which center on inclusiveness and education. Diversity, equity, and inclusion (DEI) are given top priority in its international hiring procedures, and internal learning platforms like Microsoft Learn and LinkedIn Learning aid in staff growth. Flexible work arrangements and performance rewards, which were implemented after the pandemic, encourage creativity and worker satisfaction. The business also makes investments in technical certifications and leadership development.

3- Technology Development

The secret to Microsoft's success is innovation. It makes significant R&D investments in cloud computing, quantum technology, and artificial intelligence. Its tech-forward attitude is demonstrated by its Azure OpenAI connections, GitHub Copilot, and mixed reality (HoloLens) advancements. Its software lifecycle and user experience are improved by continuous integration/deployment pipelines, cybersecurity innovations, and advanced analytics. Microsoft's alignment of innovation with environmental objectives is demonstrated by sustainability technology, such as cloud-based emissions tracking systems.

4- Procurement

At Microsoft, procurement involves finding digital infrastructure, software tools, and hardware components. Strict review procedures that prioritize sustainability, cost-effectiveness, and quality are used to govern vendor agreements. A centralized procurement system is used by the business to guarantee agility and compliance. By collaborating with ethical producers and providers of renewable energy, Microsoft improves operational effectiveness and brand credibility.

Final analysis

Microsoft's value chain exemplifies an integrated structure where sustainable advantage is driven by efficiency, innovation, and strategic alignment. Cost-effective scalability is made possible by its cloud-first strategy and software-led operations. Recurring revenue from Azure and Microsoft 365 improves long-term customer retention and financial stability. Microsoft's combination of ecosystem depth and compatibility is where it beats rivals. Cross-product adoption is promoted via smooth platform connection, ranging from Teams and Windows to Xbox and Azure. Its leadership in corporate technology is reinforced by its R&D expenditure, which powers innovative technologies like generative AI and quantum computing.

Resilience and cost management are further strengthened by supply chain agility and procurement tactics. Support services are tailored to promote diversity and creativity, particularly in HR and tech development. As further emphasized in this comprehensive Microsoft SWOT analysis of its strengths and strategic view, these actions collectively build a competitive moat founded on trust, functionality, and future-readiness. (businessmodelanalyst.com , Microsoft Value Chain Analysis, by Daniel Pereira , June 19, 2025)

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Task 2

In terms of Microsoft, Microsoft uses technology in practically every aspect of its operations. Digital tools integrate everything in the supply chain of the company, from the production of code to consumer delivery.

Platforms like GitHub and Azure serve as the digital workspaces that developers use when creating software. These solutions reduce wasted effort by enabling programmers worldwide to work together on the same projects without impacting on each other's work (microsoft.com , Microsoft 2024 Annual Report). Everything that Microsoft produces, including Windows, Office 365, and cloud services, is supported on this foundation.

Microsoft uses smart factories for physical product manufacturing. Robots are used extensively by partner manufacturers to construct Xbox consoles and Surface tablets. These automated devices can easily ramp up production when new gaming consoles launch and demand increases, and they can work quicker than human hands while maintaining constant quality.

Agile development and DevOps techniques enable the software side to operate quickly. Microsoft's ability to release security patches and updates quickly is crucial for its subscription based services. They make complex updates more dependable and seamless by using continuous integration pipelines, which automatically test and deploy code changes.

Smart technology is also essential for getting goods to consumers. By keeping copies on servers near consumers, Microsoft's Azure Content Delivery Network delivers updates and services globally. Apps load more quickly and with less delays due to this configuration (businessmodelanalyst.com , Microsoft Value Chain Analysis, by Daniel Pereira , June 19, 2025)

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Microsoft's operations are increasingly relying on sustainability tools. Their Cloud for Sustainability product reduces survey processing time by 92% by collecting emissions data from suppliers and automatically analyzing it using AI (microsoft.com , Reduce risk and improve resilience: Insights from Microsoft on advancing supply chain sustainability, April 17, 2025).

Additionally, the business employs lifecycle assessment techniques that analyze supplier materials data to monitor the environmental impact of Xbox and Surface devices (Microsoft Environmental Sustainability Report, 2024). Microsoft uses these technologies to better understand its environmental impact and fulfill its green commitments.

Microsoft can quantify and experience the real world advantages that these technologies provide. Efficiency comes up as the most evident advantage. More products are produced per hour and fewer errors are made when manufacturing is done by robots. In addition, using GitHub and Azure for all developer work reduces development time and enables clear communication between engineering teams around the world.

This tech-heavy strategy benefits Microsoft as well. Because Microsoft has more than 60 data centers across the globe, Office and Azure services are always available. This large infrastructure enables Microsoft to react quickly to any situation, whether an unexpected increase in the number of users requiring cloud services or an important security update that needs to be released immediately. These Technologies not only help in the expansion of the business but also increase its durability.

The use of technologies by Microsoft improves the customer experience. Products are being updated constantly with new features and security enhancements through the use of DevOps and agile methodologies. Chatbots and online support centers are available to assist people and the business monitors its systems to identify issues before clients see them.

In terms of Sustainability, it has both practical and reputational benefits. By automating its supplier checks and emissions tracking, Microsoft creates transparency and demonstrates to stakeholders and authorities that it takes accountability seriously. Their AI technology, which reduces supplier survey processing time by 92%, demonstrates how environmental protection and cost savings could happen (Microsoft Sustainability Blog, 2025). These actions give Microsoft the appearance of being a responsible business, which gradually increases client loyalty.

However, Microsoft also faces some challenges as a result of these technologies. The success of the company depends strongly on the outside vendors , particularly when it comes to hardware components like semiconductors. The manufacture of the Xbox and Surface was instantly impacted by the worldwide chip shortage, and as Microsoft does not produce these parts within the company, there was not much it could do about it.

Cybersecurity is also becoming increasingly complex. Hackers are attracted to Microsoft because of its extensive digital footprint, especially its cloud services. The possibility of ransomware attacks or state sponsored cyberwarfare disturbing the supply chain is increasing as more of it becomes digital.

At the top of the list of challenges is sustainability. Microsoft has improved its ability to track emissions, however, the issue is that Scope 3 emissions, or pollution from suppliers and Microsoft product users account for 97% of their carbon footprint. The company's continued construction of data centers and the rapidly growing demand for AI services are the main reasons for the approximately 26% increase in emissions in recent years (Wall Street Journal, 2025) , (Windows Central, 2025). The task of managing emissions among thousands of suppliers worldwide is considered complex.

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Task 3

According to (ebsco.com, 2023) “The first version, Windows 1.0, was released in 1985, although it faced critiques for not meeting user expectations. Significant advancements came with Windows 3.0 in 1990, which gained immense popularity, selling millions of copies and establishing Microsoft as a corporate giant”.

Task 4

Despite being one of the top giants in the industry , Microsoft could also encounter issues in its supply chain. From sourcing components to product delivery, Microsoft needs to address supply chain vulnerabilities that threaten device security at every stage of the process. Every stage has unique dangers that need to be carefully managed and watched over.

The foundation begins with the individual parts. Crucial components such as CPUs, motherboards, and chips must be obtained from vendors Microsoft can rely on and validate. Because corrupted equipment might leave security flaws that are almost impossible to repair later. due to this situation. strict quality control becomes crucial. Since external providers introduce their own risks into the mix, third-party sellers add even another level of complexity. To avoid risks like ransomware and malware from accessing their products through the back door, Microsoft must regularly analyze and approve these vendors.

Additionally, logistics presents constant difficulties. Secure handling and delivery practices help lower the risk of interception or interference because devices can be changed during shipping. Monitoring becomes essential since Microsoft must monitor components from the time of production until they are used in the field. Making vendors responsible when things go wrong and identifying security risks more quickly are made possible by maintaining accurate records.

Moreover, Microsoft may identify issues early and stop infected devices from entering the company's operations by constantly monitoring suspicious activity across the supply chain. By strengthening overall endpoint protection, this comprehensive approach to supply chain security helps lower risks even before devices are turned on. If this is done correctly, there will be fewer security issues later on and a higher guarantee that the products will reach consumers. (techcommunity.microsoft.com , feb 19,2025)

According to (techcommunity.microsoft.com , feb 19,2025) Talking about how microsoft secures the supply chain, Microsoft secures its devices from design to delivery, integrating protections at every stage of the product lifecycle. Built on the Born Secure principle, these devices incorporate security from the start—across conception, design, development, production, delivery, and maintenance. Digital signatures verify authenticity, preventing unauthorized changes or tampering.

Strict supplier oversight reinforces this approach. Microsoft conducts regular audits to mitigate top supply chain threats, including ransomware, phishing, and malware. Trusted suppliers follow strict security protocols and receive ongoing training to address evolving risks.

Software and firmware development follow Microsoft's Secure Development Lifecycle (SDL), which includes secure coding practices, digital signing, and controlled delivery. These measures verify software integrity from the moment a device boots, before Windows even loads. The SDL adapts to shifting threat landscapes and regulatory requirements.

Microsoft also applies Zero Trust principles to its engineering environment, extending protections beyond identity and access controls. Phishing-resistant multi-factor authentication (MFA) and conditional access policies help secure the development infrastructure, reinforcing security at every level.

Microsoft works beyond its own devices to strengthen global supply chain security. Participation in programs like the Customs-Trade Partnership Against Terrorism (C-TPAT) and the Transported Asset Protection Association (TAPA) reinforces protections across trade networks. C-TPAT strengthens cooperation between trade partners to prevent smuggling and terrorism, while TAPA sets global standards to safeguard high-value cargo from theft. These efforts help build a more secure and resilient technology supply chain.

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Task 5

Continuous improvements is a key component in Microsoft success. Keeping an eye on what is emerging remains Microsoft competitive in the technology industry.

> According to microsoft.com Talking about the improvement in terms of the supply chain, Advanced analytics and AI are transforming supply chain sustainability at Microsoft. By leveraging these technologies, we accelerate day-to-day work, gain real-time insights, and drive co-innovation to meet sustainability reporting requirements as well as business continuity needs. For instance, our procurement team used Microsoft Cloud for Sustainability to centralize supplier emissions data and streamline processes. By integrating AI-powered automation, such as automated review of supplier assurance letters, we reduced survey processing time by 92%.

Microsoft also utilized the six sigma concept to enhance its operations Microsoft utilized the concept of six sigma to elevate its capabilities in terms of processing improvements, waste reduction, and customer relations. (henryharvin.com , Dec 7, 2023)

In terms of process improvement at Microsoft, DMAIC adopted Process Improvement to enhance quality and minimize issues with current processes.

For customer relations, (henryharvin.com , Dec 7, 2023) talks about how Microsoft emphasizes its customer relations,

> Microsoft has greatly emphasized its customer relations in their success stories. The company walks the extra mile in not only bringing customer satisfaction but takes it to the next level of customer delight with exclamation mark AHA! by the customers.
>
> Effective customer service management needs to reduce overall costs while increasing overall customer experience and value to the customer. To bring joy to the customers Microsoft has been using the use of lean six sigma as a model for customer service management decisions.

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Task 6

Overall, Microsoft has a huge supply chain that functions effectively despite significant obstacles. The largest issue is the dependency on suppliers because essential parts like semiconductors are obtained from outside vendors. Production of the Xbox and Surface instantly slows down when shortages of chips occur.

Threats to cybersecurity are significantly more serious. Hackers are attracted to Microsoft's digital supply chain, which includes everything from GitHub to Azure data centers. Millions of users worldwide could lose access to services if a data breach happens.

Additionally, sustainability continues to decrease. 97% percent of Microsoft's carbon footprint comes from scope 3 emissions from suppliers and use of products. Data centers put a pressure on water supplies in already stressed areas, and emissions rise as demand for cloud and AI grows.

Furthermore, Microsoft is always under pressure to increase productivity and quality. Although the organization utilizes Lean and Six Sigma methodologies, it is still very difficult to manage thousands of suppliers while ensuring worldwide availability. Supply chain management is also difficult to maintain because each challenge is interconnected with the others.